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DEPARTMENT OF AVIATION & AEROSPACE MANAGEMENT

Course: Introduction to Computing Skills (Z7AVA 26209)

Laboratory Assignment: Flight Operations Analysis using Pivot Tables & Pivot Charts

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EXERCISE DATASET FILE TO USE:

05_Aviation_BLR_Airport_Pivot_Master_Dataset.csv

(Contains 50 flight records across Terminal 1 & 2 at Bengaluru Airport)

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1. SCENARIO & PROBLEM STATEMENT

You are a Junior Operations Analyst at the Airport Operations Control Centre (AOCC), Kempegowda International Airport Bengaluru (BIAL / BLR).

The Airport Director has requested an operational performance review for 50 scheduled commercial flights operated across Terminal 1 and Terminal 2.

Your objective is to provide actionable intelligence regarding:

1. Passenger distribution across airlines and terminals
2. Carrier on-time punctuality rates (OTP)
3. Root-cause turnaround delay bottlenecks
4. Peak baggage handling demands across departure waves

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2. DATASET SUMMARY (50 Flights, 13 Columns)

Col A:	Flight_ID	- Callsign/Tracking ID (FL-101 to FL-150)
Col B:	Airline	- IndiGo, Air India, Akasa Air, Vistara, SpiceJet
Col C:	Terminal	- T1 (Low-Cost Domestic), T2 (Full-Service & International)
Col D:	Sector	- Route (BLR-DEL, BLR-BOM, BLR-LHR, BLR-SFO, etc.)
Col E:	Aircraft_Type	- A320neo, A321neo, B737-MAX8, B777-300ER, B787-8, ATR-72
Col F:	Flight_Type	- Domestic vs International
Col G:	Time_Slot	- Morning, Afternoon, Evening, Night
Col H:	Pax_Capacity	- Total passenger seats installed (72 to 342)
Col I:	Pax_Booked	- Actual passengers boarded (68 to 335)
Col J:	Baggage_Pcs	- Checked-in baggage count (75 to 620)
Col K:	Status	- On-Time, Delayed, Cancelled
Col L:	Delay_Mins	- Turnaround delay duration in minutes (0 for On-Time)
Col M:	Delay_Reason	- None, Late Inbound, ATC Congestion, Snag, Weather, etc.

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3. LABORATORY TASKS (4 PIVOT TABLES + 4 PIVOT CHARTS)

WORKBOOK STRUCTURE:

- Sheet 1: Raw_Data (Imported CSV)
- Sheet 2: Task1_Pax_Terminal (Table + Clustered Column Chart)
- Sheet 3: Task2_Punctuality_OTP (Table + 100% Stacked Bar Chart)
- Sheet 4: Task3_Delay_Bottlenecks (Table + Sorted Horizontal Bar Chart)
- Sheet 5: Task4_Baggage_Waves (Table + Dual-Axis Combo Chart)

TASK 1: PASSENGER VOLUME BY AIRLINE & TERMINAL

- Rows: Airline
- Columns: Terminal
- Values: Sum of Pax_Booked (Formatted with comma separator: 1,250)
- Filter: Flight_Type
- Chart: Clustered Column Chart (Comparing T1 vs T2 per airline)
- **Question: Which airline dominates T1 passenger volume, and which dominates T2?**

TASK 2: AIRLINE ON-TIME PERFORMANCE (OTP) PUNCTUALITY

- Rows: Airline
- Columns: Status (Cancelled, Delayed, On-Time)
- Values: Count of Flight_ID
- Calculation: Show Values As -> "% of Row Total" (Percentage with 1 decimal)
- Chart: 100% Stacked Bar Chart (Green = On-Time, Amber = Delayed, Red = Cancelled)
- **Question: Which airline has the highest on-time punctuality rate?**

TASK 3: ROOT-CAUSE TURNAROUND DELAY ANALYSIS

- Rows: Delay_Reason (Filter out "None")
- Values:
  1. Average of Delay_Mins (Formatted to 1 decimal place)
  2. Count of Flight_ID (Renamed to "Affected Flights")
- Sorting: Sort rows Descending by Average Delay Minutes
- Chart: Horizontal Bar Chart with data labels enabled
- **Question: Which root cause triggers the longest average delay duration?**

TASK 4: DEPARTURE WAVES & BAGGAGE HANDLING DEMAND

- Rows: Time_Slot (Morning -> Afternoon -> Evening -> Night)
- Values:
  1. Sum of Baggage_Pcs (Renamed to "Total Baggage Pieces")
  2. Count of Flight_ID (Renamed to "Scheduled Movements")
- Chart: Combo Chart (Dual-Axis)
  - Primary Axis (Columns): Baggage Pieces
  - Secondary Axis (Line with Markers): Scheduled Movements
- **Question: Which time slot generates peak baggage carousel strain?**

4. HOW TO DO IT ON MICROSOFT EXCEL

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1. Open "05_Aviation_BLR_Airport_Pivot_Master_Dataset.csv" in Excel.

2. IMMEDIATELY Save As Excel Workbook:
File -> Save As -> Change format to "Excel Workbook (*.xlsx)".
Name: [RollNo]_[YourName]_Pivot_Assignment.xlsx

3. Rename the first sheet tab to "Raw_Data".

4. To create a Pivot Table:
Click any cell in the data (A1) -> Insert -> PivotTable -> New Worksheet.

5. In the PivotTable Fields pane on the right:
Drag fields into Rows, Columns, Values, and Filters boxes as specified.

6. Number Formatting:
Right-click any value in table -> Number Format -> Number (use 1000 separator).

7. Percentage of Row Total (Task 2):
Right-click value -> Show Values As -> "% of Row Total".

8. Changing Sum to Average (Task 3):
Click dropdown on value -> Value Field Settings -> select "Average".

9. Inserting Charts:
Click inside Pivot Table -> PivotTable Analyze -> PivotChart -> Select Chart Type.

10. Combo Chart (Task 4):
PivotTable Analyze -> PivotChart -> Combo -> Set Baggage to Column,
Flights to Line and check "Secondary Axis".

5. HOW TO DO IT ON GOOGLE SHEETS

STEP 1: IMPORT CSV INTO GOOGLE SHEETS (MANDATORY)

1. Open browser -> Go to https://sheets.new (opens a new blank sheet).
2. Go to top menu: File -> Import.
3. Click the "Upload" tab on the right.
4. Drag and drop "05_Aviation_BLR_Airport_Pivot_Master_Dataset.csv" (or Browse).
5. In the Import Location dialog:
   - Import location: "Replace current sheet"
   - Separator type: "Detect automatically"
   - Convert text to numbers, dates, and formulas: Check "Yes"
6. Click "Import data".
7. Rename bottom-left sheet tab to "Raw_Data".
8. Name spreadsheet at top left: [RollNo]_[YourName]_Aviation_Pivot_Lab.

STEP 2: CREATE PIVOT TABLES & CHARTS

1. Click cell A1 -> Press Ctrl + A to select all data.
2. Top menu: Insert -> Pivot table -> Select "New sheet" -> Click "Create".
3. Rename tab to "Task1_Pax_Terminal".
4. In the Pivot Table Editor sidebar on the right:
   - Rows: Click "Add" -> select field.
   - Columns: Click "Add" -> select field.
   - Values: Click "Add" -> select field -> choose Summarize by (SUM, COUNTA, AVERAGE).
   - Filters: Click "Add" -> select field.
5. Show as Percentage of Row (Task 2):
   In Values section -> change "Show as" from Default to "% of row".
6. Insert Charts:
   Highlight the Pivot Table range -> Insert -> Chart.
   In Chart Editor (Setup tab): Choose Chart Type (Column, 100% Stacked Bar, Bar, Combo).
   In Chart Editor (Customize tab): Add Chart Titles, Legend, and Data Labels.

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6. SUBMISSION GUIDELINES

- **DEADLINE: 26 October, 2026 | 1000 hours**
- **FILE FORMAT: All submissions must be in Excel (.xlsx) format.**
  * Excel users: Save your file as [RollNo]_[YourName]_Pivot_Assignment.xlsx
  * Google Sheets users: File -> Download -> Microsoft Excel (.xlsx)

- **UPLOAD LINK (Google Drive):**
  https://drive.google.com/drive/folders/1pPt5eKaWxiPeLtm0UoopuW3rwkjRVTAV?usp=drive_link

- EVALUATION: 50 Marks Total (10 marks per task + 5 marks data setup + 5 marks analysis).

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